Twist-assisted IC: experiment program

Date: 2026-10-21. Scope: pursue routes 1–3 and their combinations, devise discriminating experiments, and preserve adjacent conditional routes without claiming an algorithm or novelty.

This program extends PR #2380 and the central tracker. It does not substitute an unexecuted plan for measured results. Bounded native panels have now run; the execution results separate verified finite checks, partial measurements and remaining construction/validation gaps. The original acceptance criteria below are retained.

Questions and deliverables

Requested item	Deliverable	Status
Pursue trace restrictions	Explicit coordinate, slope, norm, multiplicative and divisor families; E01–E06	Bounded native restrictions/censuses executed; broader full-cost criteria open
Pursue covers	Explicit C_c family and complement equations, gluing alternatives, subgroup/class-map gates; E10–E13	Ten executable class transfers checked; destination solver/full cost open
Pursue solver symmetry	Four membership encodings, equivalent rational witnesses, torsion descent and solver panel; E07–E09	Equivalent systems and three bounded solver attempts executed; broader grid/full cost open
Pursue combinations	Representation/isogeny ablations and full factorial pipeline panel; E14–E15	1,920 native factorial records; 1,704 exact verified replays; cost calibration open
Preserve other angles	Exact pairing applicability and bounded characteristic-zero lifting ingress; E16–E17	Public applicability and bounded lift/pairing controls executed; no NIST transfer supplied
Make formula checks reproducible	Independent small Rust crate reusing the repository identity checker	E00 and nine explicit Lean lemmas checked; concrete field/group/map bridges remain open
Preserve follow-up execution	Three versioned backlog protocols, machine-readable experiment matrix, graph and report PDF	Results and receipts recorded; cards remain blocked for unmet full-campaign criteria

Common mathematical contract

The source remains a declared E/F_p, characteristic p > 3. The actual DLP target is the original rational order-r subgroup, not the larger ambient E(F_(p²)) group. An x-preserving twist uses d*y²=f(x), omega²=d a nonsquare. Alternative models carry their explicit conversion.

The trace is tau=1+pi. Anti-invariant points lie in its kernel. Over an odd-order rational group, a trace fiber is [2^-1]T+E^-; counting all lifts can inflate witness counts while leaving projected rows unchanged. This is the reason for the exact-F and rational-witness controls, not a complexity lower bound against all coordinate algorithms.

Concrete trace families

Write S=(u+v*omega*,*w*+*z*omega).

- **Coordinate boxes:** u in U_H and v in V_H, with U_H=V_H={0,...,H-1} and H in {2,4,8}; both signs of y and all exceptional branches retained.

- **Slope boxes:** u in U_H and m=z/v in M_H, with the v!=0 branch certified separately. The trace has X=m²-2u. A pure-F_p witness formulation uses the same quartic below, without representing an extension point.
- **Norm families:** u²-dv² in W_H, optionally intersected with a coordinate box; analogous y-norm restrictions are separate arms.
- **Multiplicative families:** subgroups/power images only at H dividing p-1, with the exact image size checked. A parameterization with costly membership is not a free small base.
- **Divisor families:** generic degree-two places versus pure twist pairs. Principal divisors of x-x0 are removed before row formation; they carry no new class relation.

The quartic used in the slope arm is

$$\Delta_T(m) = m^4 - 6Xm^2 - 8Ym - 3X^2 - 4A, \qquad 4dv^2 = \Delta_T(m).$$

The line-intersection identity gives a rational-only formulation with exactly the same witnesses. E00 checks the polynomial arithmetic; the group interpretation and exceptional cases remain separate obligations.

A concrete cover family and its complement

Let f(x)=x³+Ax+B and choose c with f(c)!=0. Consider

$$C_c : \; v^2 = f(u^2 + c).$$

The two quotient maps on their affine domains are

$$(u, v) \longmapsto (u^2 + c, v) \in E, \qquad (u, v) \longmapsto (t, z) = (u^2, uv) \in D_c,$$

where

$$D_c : \; z^2 = tf(t + c).$$

For t!=0, put F=f(c), a_2=3c²+A and

$$X = F/t + a_2/3, \qquad Y = Fz/t^2.$$

The short complement coefficients are represented without denominators by

$$3A_D = 3Ac^2 + 9Bc - A^2,$$

$$27B_D = -27Bc^3 + 18A^2c^2 + 27ABc + 27B^2 + 2A^3.$$

These are candidate explicit model-conversion formulas with a polynomial preflight, not a certified Jacobian subgroup transfer. Require nonsingularity, f(c)!=0, characteristic >3, t!=0 on this affine chart, separate infinity/ramification charts, and actual divisor-class maps.

The complement must be checked: compare its trace and j, then construct an F_p-isomorphism to the requested twist when claimed. Same j alone confuses twists; opposite trace alone supplies no explicit map. Preserve complements that fail the twist test as controls.

For a smooth projective geometrically irreducible C/F_p, genus g, with a nonconstant degree-delta map to E, the standard point/fiber bound gives an order-p image when g and delta are fixed. E13 therefore tests restrictions and pays for them; embedding the original subgroup in a larger Jacobian is not itself an advantage.

Inputs and selection rules

The already registered ordinary prime-order structural fixtures are:

Source curve	p	A	B	#E(F_p)
icv1-fp6-tm7-fc20da94	59	1	13	67
icv1-fp10-t5-192cb216	827	1	15	823

They test group shape and implementation mechanics, not NIST coefficient fidelity or scaling. The registered even-order controls icv1-fp7-t4-0c09e92f and icv1-fp7-t2-4910f2d0 expose 2-primary assumptions; the odd-composite control icv1-fp8-tm1-1ed3feb7 separates oddness from primality. Verify all parameters and subgroup choices rather than importing a generator by bit count.

For later scaling, the proposed train/holdout ladder has prime-field bit lengths {8,10,12,14}, A=-3, ordinary odd prime-order curves, and two independently generated instances per rung. Freeze the generator, all rejected candidates, registered ICV1/EC1 representations, extension moduli, generators, exact fixture files and SHA-256 hashes in an amendment before timing. No such larger fixtures have been generated or admitted by this plan.

Stage panels use H in {2,4,8}, arity m in {2,3,4}, projected base budgets B in {8,12,16}, and declared seeds 0–3 for design and 4–7 for holdout. Unsupported and empty cells stay explicit. The resource envelope and expected cost need calibration before readiness; these grids are design inputs, not an authorization to launch a paid campaign.

Tiny exact-census cells enumerate every nonidentity rational target plus a separate identity branch. A performance cell is one previously unseen public target, with target generation excluded symmetrically from the online intervals. Planted decompositions are correctness controls only. Do not use their yield as a uniform-target estimate.

Experiment matrix

The canonical machine-readable version is experiments.json. Every row below names a falsifier and what its outcome would change.

E00 — Polynomial preflight

- **Route / status:** common / verified polynomial preflight and five Lean polynomial lemmas; concrete group/map bridges remain open.
- **Question:** Do the proposed trace and complement coordinate formulas satisfy their exact cleared-denominator polynomial identities?
- **Arms and inputs:** Five compiled statement ASTs; exact integer coefficient expansion and repository identity certificates; deliberately false +1 variants.
- **Controls:** Exact expansion and modular evaluation are different arithmetic checks; all negative variants must reject.
- **Measurements / acceptance:** All five exact coefficient differences zero; 32 issuer and 32 domain-separated checker points pass; five false variants reject.
- **Stop, null outcome and next decision:** Any mismatch stops use of the affected formula; no transfer or performance claim follows from passing.
- **Prerequisites:** source and E00 protocol frozen before execution.

E01 — Trace fibers and separation

- **Route / status:** 1 / verified exhaustive bounded fixture census; general formal bridge open.
- **Question:** Which apparent relation gains are just multiple lifts of the same trace?
- **Arms and inputs:** All points on the two registered odd-prime-order toy curves over the declared quadratic extension; add separate registered even-order and odd-composite controls.
- **Controls:** Rational-only traces; pure twist trace zero; S+U with U in ker(tau); identity; rational 2-primary cases.

- **Measurements / acceptance:** Exact trace image/fiber counts, distinct projected rows, row ranks and target-row span.
- **Stop, null outcome and next decision:** Do not count repeated fibers as new information; a failed asserted identity is a correctness failure, not a performance loss.
- **Prerequisites:** registered exact extension models and native point verifier.

E02 — Coordinate boxes

- **Route / status:** 1 / partial: declared restrictions executed; full larger-field norm-y/power families open.
- **Question:** Do small coordinate boxes create a useful small trace base with cheaper membership or decomposition?
- **Arms and inputs:** S=(u+v omega,w+z omega) with u in U and v in V; H in {2,4,8}; enumerate both y signs and branches.
- **Controls:** Exact-image rational base; same-size hash-selected rational base; unrestricted lift control.
- **Measurements / acceptance:** Distinct traces, nonzero coverage, fiber multiplicities, construction/encoding cost and verified independent rows.
- **Stop, null outcome and next decision:** Record empty bases and size collisions; never normalize away construction or duplicate-removal work.
- **Prerequisites:** E01.

E03 — Slope and real-part restrictions

- **Route / status:** 1+3 / verified 30 finite witness-set comparisons; cost comparison open.
- **Question:** Does restricting u and m=z/v help beyond a rational parametrization of X=m^2-2u?
- **Arms and inputs:** u in U and m in M, H in {2,4,8}; enforce 4dv^2=Delta_T(m); add independent v restriction as an ablation.
- **Controls:** Same quartic witness written entirely over Fp; exact-F table; ordinary low-degree addition witnesses.
- **Measurements / acceptance:** Projected solution-set equality first; then encoding size and counted work per independent row.
- **Stop, null outcome and next decision:** If the candidate and rational witness systems are identical after renaming, classify the result as encoding reuse rather than new twist geometry.
- **Prerequisites:** E01.

E04 — Norm and multiplicative restrictions

- **Route / status:** 1 / partial: norm/image controls executed; larger-field fibers and costing open.
- **Question:** Can norm constraints or multiplicative structure reduce solve cost after parameterization and membership are charged?
- **Arms and inputs:** Norm(x)=u^{2-dv}2 in W; norm(y) restrictions; H-th-power or subgroup conditions only when H divides p-1; intersect with a trace-box family.
- **Controls:** Same F without norm witnesses; same-size rational base; matched subgroup-size control; nonsquare d rescaling.
- **Measurements / acceptance:** Image size, membership degree/selector size, projection collisions and complete stage work.
- **Stop, null outcome and next decision:** Unchanged projected sets at unchanged or higher cost close only that restriction; missing roots do not count as a win.
- **Prerequisites:** E01.

E05 — Divisors and degree-two places

- **Route / status:** 1+2 / partial: two sparse-function/divisor controls; unsplit degree 3/4 factors open.
- **Question:** Can generic conjugate pairs help smooth-divisor relations when the principal pure-twist pairs are removed?

- **Arms and inputs:** Sparse rational functions a(x)+y b(x), coefficient support frozen; distinguish x in Fp from genuinely nonrational degree-two places.
- **Controls:** Remove divisors of x-x0 before forming rows; rational-only function family; generic degree-two-place control.
- **Measurements / acceptance:** Factoring and lifting work, nontrivial projected divisor rows, row span and failed factorizations.
- **Stop, null outcome and next decision:** Principal pairs and empty projected rows are not relations; retain the full degree and coefficient-height budgets.
- **Prerequisites:** E01; native divisor and factorization verifier.

E06 — Exact relation census

- **Route / status:** 1+3 / partial: 1179 configured census rows, 678 verified and 501 unsupported; decomposition cost open.
- **Question:** Which restricted bases actually span the needed target rows?
- **Arms and inputs:** m in {2,3,4}, B in {8,12,16}; all nonidentity targets for tiny fixtures; freeze sign/orbit conventions.
- **Controls:** Complete enumeration or MITM oracle; identical projected base and target; raw versus canonicalized rows.
- **Measurements / acceptance:** Exact relation set, target coverage, rank over r, target-row span and verified known-answer recovery.
- **Stop, null outcome and next decision:** Do not silently replace unattainable full-column rank with a new stopping rule; freeze target-span stopping in advance.
- **Prerequisites:** E02 or E03 or E04.

E07 — Equivalent system builders

- **Route / status:** 3 / partial: 240 distinct B8/H8 projection cells verified; B12/16 and other bases open.
- **Question:** Which membership representation avoids merely moving cost into a higher-degree constraint?
- **Arms and inputs:** Univariate product membership, bit decomposition with proven range, selector/product chains and explicit branching.
- **Controls:** Untwisted addition chain; exact same projected F; independently enumerated projected solutions.
- **Measurements / acceptance:** Equality of full projected sets including multiplicity/sign policy and exceptional branches; equations/variables/degrees.
- **Stop, null outcome and next decision:** A mismatch blocks all solver comparisons until repaired; coefficient fields and modular-wrap conditions are part of the input.
- **Prerequisites:** E06.

E08 — Solver and ordering panel

- **Route / status:** 3 / partial: three 1930-cell bounded solver attempts retained; applicable signature reference, broader grid and full costs open.
- **Question:** Does symmetry reduce actual solving cost against the strongest equivalent encoding?
- **Arms and inputs:** Frozen variable orders, chain/quadratized/symmetric encodings; msolve/open reference plus available applicable F4/F5 engines.
- **Controls:** A/A runs; untwisted witness system; redundant-variable positive/negative controls; same output verifier.
- **Measurements / acceptance:** Counted native operations, Macaulay dimensions/nnz, reductions, solving degree, memory, failures and cost per useful row.
- **Stop, null outcome and next decision:** Lower displayed degree alone is not a win; timeouts and OOMs remain incomplete outcomes under the same envelope.
- **Prerequisites:** E07; calibrated external-solver adapters.

E09 — Torsion over extensions

- **Route / status:** 1+3 / verified bounded translated-torsion controls; general proof bridge open.
- **Question:** Can torsion symmetry survive Galois descent without false factor-base compression?
- **Arms and inputs:** 2-torsion over Fp^3 and combined Fp^6 where applicable; Galois-invariant coordinates and descent constraints.
- **Controls:** No-torsion encoding; [2] map on the odd subgroup; retained extension-field cost; rational torsion special-family control.
- **Measurements / acceptance:** Equivalence, actual orbit/image size, descended system degree and complete representation cost.
- **Stop, null outcome and next decision:** A geometric torsion orbit that leaves the rational base is not a rational factor-base quotient.
- **Prerequisites:** E07; registered extension models and torsion verification.

E10 — Explicit degree-two covers

- **Route / status:** 2 / verified exhaustive declared-family search over 27140 parameters on ten fixtures.
- **Question:** When is the complementary elliptic quotient actually the required twist?
- **Arms and inputs:** C_c: $v^{2=f(u}$2+c), f(c)!=0; quotient E and D_c: z^2=t f(t+c); use the explicit short complement coefficients.
- **Controls:** Reject f(c)=0; same-j only control; trace-opposite only control; explicit Fp isomorphism check.
- **Measurements / acceptance:** Smoothness/genus, complete quotient maps including infinity, complement trace/j, isomorphism and image size.
- **Stop, null outcome and next decision:** Neither matching j nor opposite trace alone certifies the requested isomorphism or subgroup transfer.
- **Prerequisites:** E00; native smoothness and map checker.

E11 — Gluing and alternative complements

- **Route / status:** 2 / partial: module filters and two-cover search; three positive toy module cases and the public P-521 module filter need actual torsion/polarization/quotient/map constructions.
- **Question:** Do other gluing orders or complementary-factor constructions provide a better executable route?
- **Arms and inputs:** Galois-equivariant 2/3-torsion gluing where applicable; recover the complement from a supplied cover; bounded typed candidate families.
- **Controls:** Degenerate/product polarization control; incompatible Frobenius control; identical-field checks.
- **Measurements / acceptance:** Actual curve/map construction, indecomposable polarization, subgroup kernel, factored evaluation cost and failures.
- **Stop, null outcome and next decision:** A rejected family or failed bounded search does not establish universal nonexistence.
- **Prerequisites:** E10; gluing/complement implementation.

E12 — Jacobian arithmetic and recovery

- **Route / status:** 2 / partial: ten executable transfers, 5855 classes, independent PARI orders; formal/independent-host class-map bridge open.
- **Question:** Does the proposed transfer work on the divisor classes the collector actually produces?
- **Arms and inputs:** Pullback/norm on the chosen infinity model; arbitrary valid reduced classes; image subgroup order and recovery.
- **Controls:** phi_*phi*^=[degree]; origin and ramified fibers; independent CAS; direct-E solver.
- **Measurements / acceptance:** Exact group-law and kernel/recovery checks; separate setup, maps, arithmetic and recovery costs.
- **Stop, null outcome and next decision:** Unsupported two-infinity or arbitrary-class arithmetic remains unsupported, not approximated by rational point-class tests.
- **Prerequisites:** E10 or E11; native divisor-class arithmetic and norm.

E13 — Restricted cover images

- **Route / status:** 1+2 / partial: low-height image census; Jacobian decomposition and useful-row cost open.
- **Question:** Can restricting cover points give a small base without a worse membership or map problem?
- **Arms and inputs:** All-C(Fp) control; restricted u/trace/norm supports; degree/genus ladder only with frozen factored maps.
- **Controls:** Same projected base on E; fixed-genus image bound; independent point counts.
- **Measurements / acceptance:** Actual image size, g/degree/representation size, construction and membership cost, projected rows.
- **Stop, null outcome and next decision:** A fixed bounded-degree cover with order-p image does not supply a small base merely by changing the ambient variety.
- **Prerequisites:** E12.

E14 — Model and isogeny ablations

- **Route / status:** 1+2+3 / verified bounded map/dual construction: three explicit degree-seven maps and all three algebraic duals in V2; full IC costs open.
- **Question:** Are any savings due to cheaper constants or another isogenous representative rather than new relations?
- **Arms and inputs:** Nonsquare d choices, x-preserving/standard twist models, fixed small-degree rational isogeny routes and verified return maps.
- **Controls:** Equivalent direct-E representation; same source target/subgroup; conversion charged both ways.
- **Measurements / acceptance:** Identical recovered outputs and projected relation policy; map, primitive and solver cost separated.
- **Stop, null outcome and next decision:** Different j or field models require typed identities; do not infer an unknown-scalar speedup from faster known-scalar multiplication.
- **Prerequisites:** E07; verified map adapters.

E15 — Factorial pipeline ablation

- **Route / status:** 1+2+3 / partial: 1920 native records, 1704 verified and identically replayed, 216 empty-base errors; calibrated full cost and independent host open.
- **Question:** Which component or combination improves the actual IC baseline?
- **Arms and inputs:** Trace restriction on/off, cover on/off, solver symmetry on/off; pairs use equal target/base or explicitly separate changed-base studies.
- **Controls:** Strongest current IC baseline; strongest same-target rho; direct destination generic solver; independent holdouts.
- **Measurements / acceptance:** Cold single-target complete counted cost and S; setup/online split; 0/0.1/1 GAE pricing band; verification and replay.
- **Stop, null outcome and next decision:** Every claimed IC improvement needs baseline/candidate >1 with all phases; rho is a reported gap, not the IC improvement pass gate.
- **Prerequisites:** E06; E08; E12 for cover cells.

E16 — Homomorphism and pairing applicability

- **Route / status:** conditional / verified fifteen public-parameter checks and known degree-two pairing positive control; no NIST transfer supplied.
- **Question:** Do small twist factors or a pairing actually supply an admissible transfer on the original subgroup?
- **Arms and inputs:** Exact gcd(r,#E_tw), embedding-degree and torsion/Frobenius tests on the declared public parameters.
- **Controls:** Trace/norm killing pure anti-invariants; translated-coset control; known small-embedding-degree positive family kept separate.
- **Measurements / acceptance:** Exact applicability certificates and a named nondegenerate transfer if one exists.

- **Stop, null outcome and next decision:** A smooth twist order or shared x-coordinate relation does not supply a subgroup-preserving transfer; do not run a nonexistent destination attack.
- **Prerequisites:** exact public-parameter and transfer checker.

E17 — Arithmetic lift and twist ingress

- **Route / status:** conditional / partial: integral and two 254-case rational-height panels; one rank interval, target lifting and discovery cost open.
- **Question:** Could a characteristic-zero twist reduce to E while providing a usable small-height relation source?
- **Arms and inputs:** Only twists whose parameter is square mod p for a same-E reduction; bounded height/rank/search families stated before a pilot.
- **Controls:** Nonsquare reduction goes to the other finite-field component; fixed-lift and equal-height controls; charge discovery and target lifting.
- **Measurements / acceptance:** Explicit reduction map, target lift, relation independence, heights and construction/search costs before any solver comparison.
- **Stop, null outcome and next decision:** High rank alone is not a cheap target lift or a speedup; no unbounded rank or designer-secret assumption is treated as evidence.
- **Prerequisites:** separate arithmetic-lift construction and height protocol.

Fair comparisons and attribution

Run three controls when a candidate provides an image F:

1. **Exact-F table:** materialize the identical projected point set and charge construction/storage.
2. **Rational witness:** retain every useful F_p auxiliary parameter and the same quartic/membership constraints; remove only the extension-field interpretation.
3. **Strong ordinary IC baseline:** the best applicable verified pipeline with its own fair tuning budget, plus a matched strong rho reference.

A candidate losing its apparent advantage against control 2 has not discovered a benefit from twist relations. It may still be a useful encoding optimization, which is reported against the previous IC baseline. If a cover changes F, either compare paired arms on the same F or declare a separate changed-base experiment; an eight-cell factorial design cannot assume unrelated workloads match.

Membership is part of the problem. Compare product polynomials, bounded bit decompositions, selector/product chains and explicit branch enumeration with all selection, range and modular-wrap checks charged. A high-degree membership constraint cannot be omitted from the solver input.

Use canonical signed/orbit representatives only under a proved rule. Count projected nontrivial coefficient rows, their rank and the target-row span. Duplicate lifts, zero rows and scalar/multiple rows stay in the raw evidence and are not called independent relations. Full column rank is not silently assumed; target recoverability and the stopping rule are frozen before execution.

Success, stopping and cost

The immediate success criterion is a verified lower cost against the best current IC baseline on identical accepted workloads. For a stage, compare counted cost per independent useful projected row and label it a stage result. For a full pipeline, report baseline_total/candidate_total >1 with every phase included. A rho crossover is a separate claim requiring the same target, complete intervals and the strongest applicable rho implementation.

Use the repository's single-target ecbench contract for method comparisons. Report cold cost, setup and online costs separately; retain attempted and completed counts, verification, failed searches, OOMs and timeouts. Price uncalibrated work at 0, 0.1 and 1 GAE symmetrically for all arms. A ranking that changes within this band is unresolved. Wall-time results additionally require isolation, A/A controls, paired reruns and admitted uncertainty.

For solver experiments, export exactly the same systems and verify full projected solution sets before ranking. Compare with the fastest available applicable external implementation; missing Magma or

other proprietary comparisons are named, not silently replaced by a weaker reference. Exponents require at least four sizes and full phase costs; two toy fixtures are not a scaling result.

For an admitted performance comparison, freeze per-cell limits after cost calibration. The initial L0 diagnostic caps were frozen before execution, with full calibration still open and no speed claim admitted. Stop the affected cell on a correctness mismatch. Preserve budget exhaustion as an incomplete outcome under the original envelope. A null result closes the tested family only, not every possible twist-related construction.

Dependency and execution order

- E00 can run after its source and protocol are committed.
- E01 and E10 are independent construction/geometry diagnostics.
- E02–E06 generate and qualify restricted bases; E07 verifies system equivalence before E08/E09 timing.
- E11/E12 must establish actual divisor-class maps before E13 or any cover-containing E15 cell.
- E14 separates representation costs from relation effects.
- E15 is the final combined pipeline panel; E16/E17 remain conditional ingress tests with no assumed transfer.
- Backlog route cards own these phases and remain blocked until their actual native drivers/checkers, source pins, resource estimates and fixture hashes exist. No unsupported component is marked runnable by adding this document.

See PROGRAM.svg for the dependency diagram.

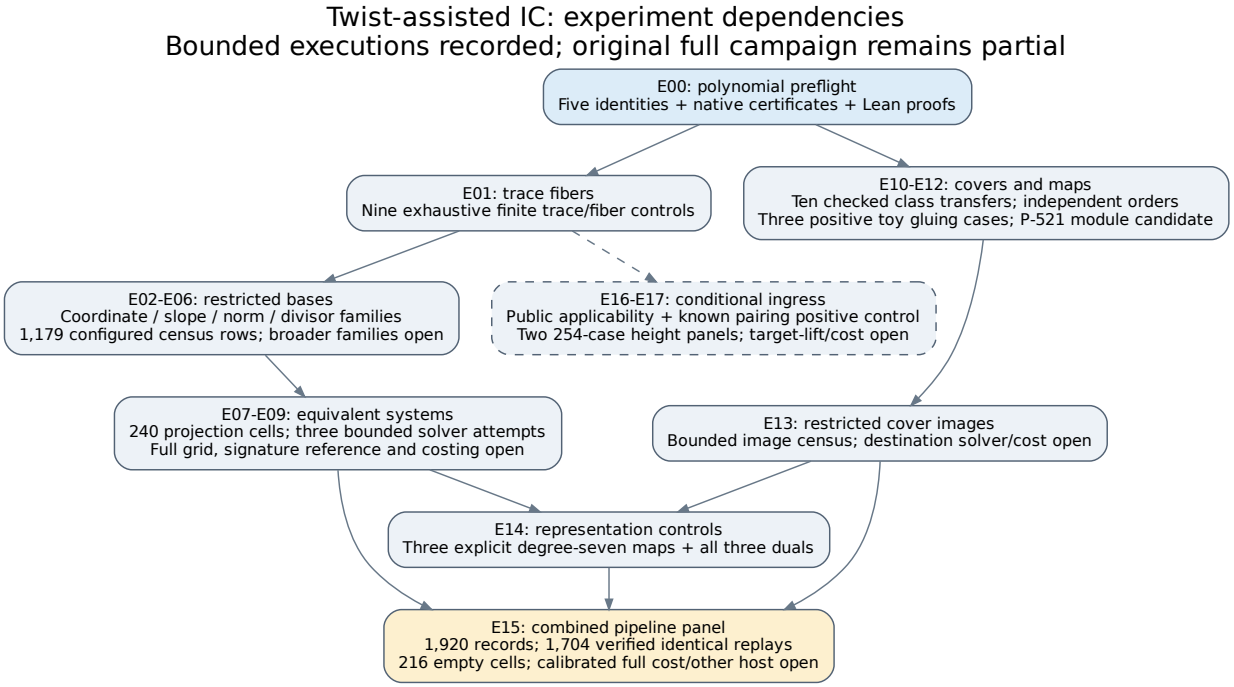


Figure 1: Experiment dependencies and bounded execution; the original full campaign remains partial

The PDF includes that same diagram. No performance dashboard value changes until a result meets the repository validation gate.

Proof map and limits

Item	Check implemented here	Remaining proof / execution obligation
Chord polynomial division and quartic expansion	E00 coefficient/evaluated certificates and corresponding Lean identities	Concrete elliptic group and exceptional-chart bridge
Complement coefficients and cleared model conversion	E00 arithmetic, Lean identities and ten executable toy class transfers	Faithful formal field/chart/Jacobian maps and independent arbitrary-class replay
Trace-fiber and separation statements	Nine exhaustive native censuses and four conditional Weierstrass point-group Lean statements	Concrete finite-field Frobenius/twist and odd-subgroup bridge
Image-size bound	A stated standard bound with explicit hypotheses	Applicable curve/morphism hypotheses and proof mapping for any selected candidate
Solver equivalence and benefit	240 distinct projection checks, bounded solver panels and native IC/replay	Broader grids, complete applicable external reference and calibrated full costs
NIST-size applicability or novelty	Not established	Separate scalable construction, applicable evidence and a literature novelty audit

A certificate with proof=null is evaluated-tier, not a Lean proof. Different checker points in one process do not constitute independent-host replay. Clearing denominators certifies only a polynomial identity; it does not prove the original rational map is defined everywhere.

Sources

- Semaev, summation polynomials, ePrint 2004/031.
- Faugere–Huot–Joux–Renault–Vitse, torsion symmetries, EUROCRYPT 2014.
- Joux–Vitse, cover and decomposition, ePrint 2011/020; extension-field results are not a NIST prime-field result.
- Broker–Howe–Lauter–Stevenhagen, gluing constructions.
- Gallese–Lombardo–Naccarato–Zannier, Finding the complement of an elliptic curve inside a Jacobian, arXiv:2606.02429v1. Prior art for computing a complementary factor, not an IC speedup.
- Higher-genus implementation contract, identity certificate contract.
- PR #2368, unmerged prior mixed-component study at the time this program was written; no new replay of its results is claimed here.

Execution evidence

The historical E00 preflight report checks five exact polynomial identities and evaluated certificates, rejects five false variants, and records 22 supporting tests. The subsequent bounded execution report and per-experiment statuses above record E01–E17 execution/applicability evidence and every remaining original requirement. No calibrated speedup or novelty claim follows.